STANDARDIZING PUBLIC RAT RNA-SEQ DATA THROUGH METADATA CURATION AND REPROCESSING

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https://rgd.mcw.edu

Rat Genome Database

Abstract

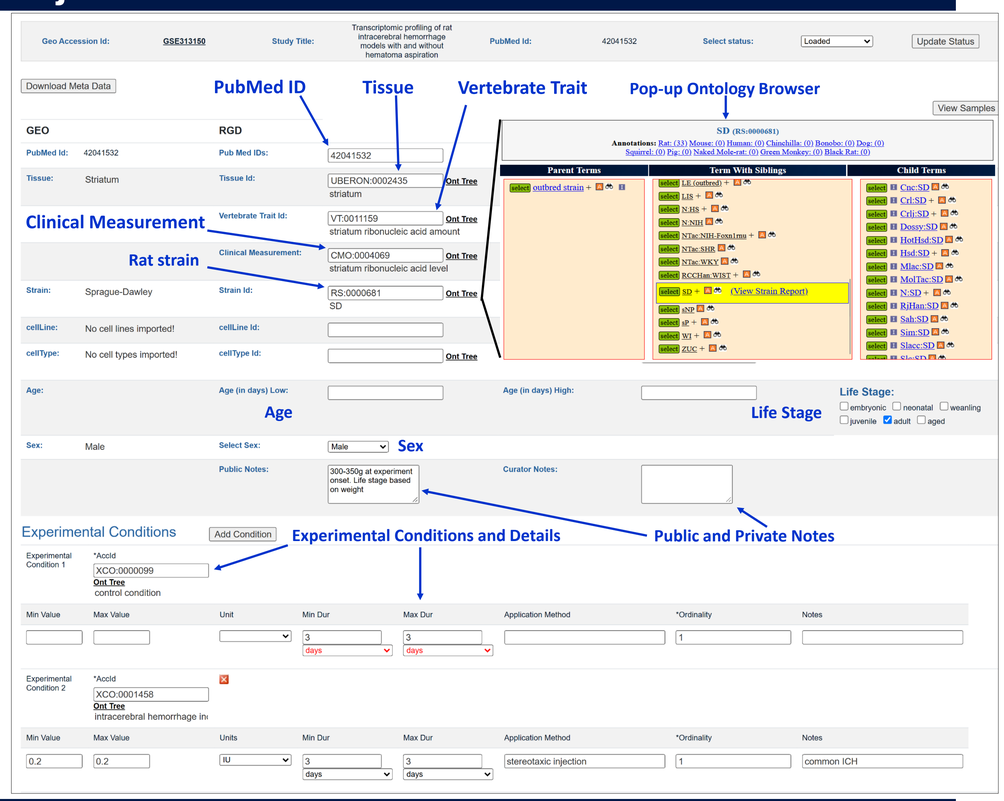
High-throughput RNA sequencing (RNA-seq) is a valuable tool for studying gene function, disease, and physiology, with the volume of datasets in public repositories steadily increasing. These public repositories contain extensive collections of transcriptomic datasets; however, the study descriptions are often incomplete or missing contextual information needed for broad reuse by the research community. As the NIH-designated resource for *Rattus norvegicus* (rat), the Rat Genome Database (RGD; https://rgd.mcw.edu) is well positioned to integrate publicly available rat gene expression data with curated rat-specific biological knowledge. To expand the utility of public rat RNA-seq datasets, RGD has developed an integrated framework for acquisition, curation, processing, and dissemination of expression data deposited in the Gene Expression Omnibus (GEO).

Here we describe the recent expansion of RGD's RNA-seq content through the development of an NLP-assisted, ontology-based metadata curation workflow that improves study searchability and accessibility. To further support data exploration and reuse, RGD has standardized quantitative expression data by reprocessing associated RNA-seq FASTQ files through a bioinformatic pipeline that aligns sequencing reads to the GRCr8 rat reference genome assembly. This standardization establishes a robust framework for cross-study comparisons while also providing data quality snapshots, including raw read metrics, alignment statistics, species verification, inferred sample sex, and normalized coverage profile visualizations. Metadata quality is enhanced through ontology-based standardization, identification of missing study information, and the resolution of ambiguities through literature review and communication with data submitters and authors.

Together, these efforts reduce barriers to cross-study comparisons of publicly available rat RNA-seq datasets by applying a consistent analytical and curation framework to studies representing over two decades of transcriptomic research. The integration of curated metadata, standardized expression values, and interactive visualizations throughout the RGD ecosystem creates a centralized resource for the rat research community, improving the accessibility, consistency, and utility of publicly available rat transcriptomic data.

Objective 1: Metadata Standardization

Figure 1. Metadata are extracted from the imported GEO study SOFT files. The NLP algorithm auto-populates tissue (UBERON), cell type (CL), and strain (RS) terms. Vertebrate trait (VT) and clinical measurement (CMO) terms are assigned automatically from mappings to the UBERON term. Sex and PubMed ID are parsed directly from the file when stated. Age, when provided in an appropriately structured field, is displayed in the interface and reviewed prior to being entered by the curator. Experimental conditions (XCO) and their details are assigned manually. Curators add, accept, or override terms as needed. Pop-up ontology browsers (inset) are available for each field. Studies are first curated at the study level and terms are propagated to all applicable samples.



Objective 2: Quantitative Expression Data Standardization

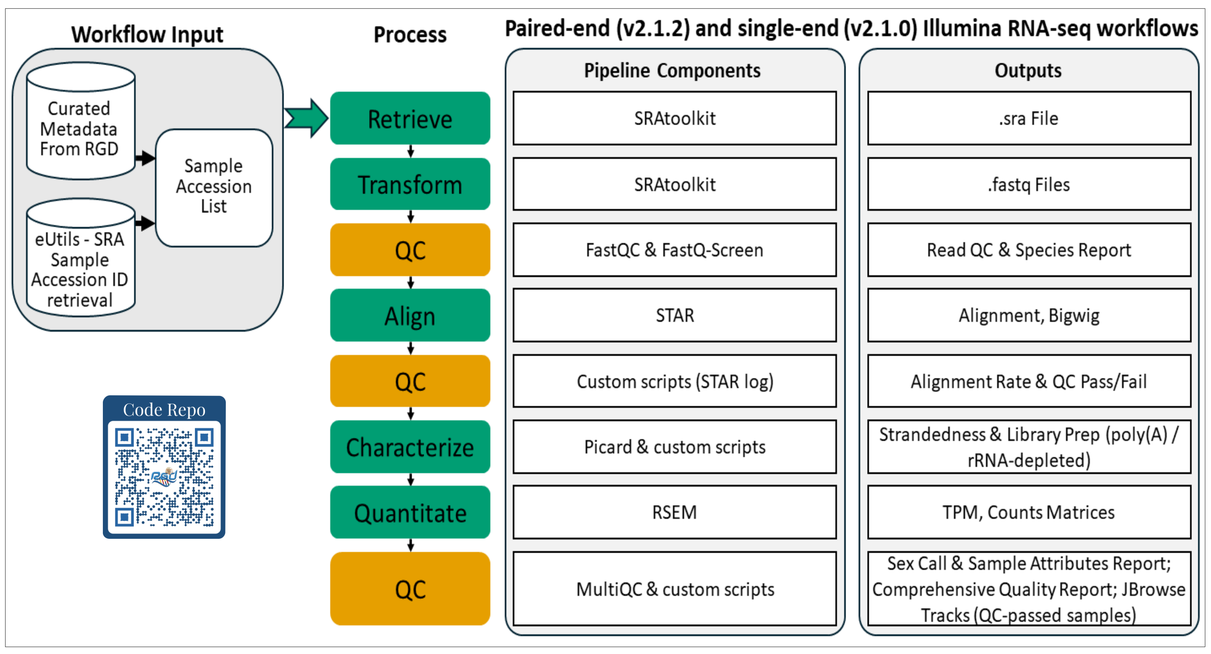


Figure 2. The RGD RNA-seq reprocessing pipelines use publicly available software to retrieve and process raw Illumina reads from the SRA. STAR performs the alignment and RSEM estimates expression abundance. The pipelines also produce outputs that GEO submissions often lack: species verification, read and alignment quality reports, BigWig coverage files, and computed sex. These give users more insight into each dataset. Since version 2.1, both pipelines assign sample sex using a codified rule and also compute library strandedness and preparation method.

Objective 3: Data Access in RGD - Gene Report

Cyp1a1 Gene Report

Figure 3. Exploring rat *Cyp1a1* expression.
A) In the Gene Report's RNA-SEQ Expression section, selecting Respiratory System shows all records with TPM > 0.
B) Filter by TPM level or sort by any metadata field; GEO-reported sex is shown beside computed sex.
C) Links to Strain/Cell Line, Reference, and RGD Study Report.
D) Export all or selected records to .csv.

Figure 4 follows study GSE162239.

Gene: Cyp1a1 (cytochrome P450, family 1 subfamily a, polypeptide 1) Rattus norvegicus

RNA-SEQ Expression

Click on a value in the shaded box below the category label to view a detailed expression data table for that system.

Download All Expressed Objects for this Gene
Download Selected Expressed Objects

3D

adipose tissue	alimentary part of gastrointestinal system	appendage	circulatory system	ectoderm	endocrine system	endoderm	entire extraembryonic component	exocrine system	hemolymphoid system	hepatobiliary system	integumental system	mesenchyme	mesoderm	musculoskeletal system	nervous system	pharyngeal arch	renal system	reproductive system	respiratory system	sensory
282	259	635	1420	3110	1501	2446	130	1341	137	1327	207	5774	2084	170	2886	27	1903	674	1141	216

3A

Hide Table

Filter by Expression Level: High: TPM > 1000 Medium: 10 < TPM ≤ 1000 TPM Low: 0.5 ≤ TPM ≤ 10 Below Cutoff: TPM < 0.5 Clear Filters Showing 1141 of 1141 records

3B

Strain/ CellLine	Sex	Computed Sex	Age	Tissue	Source Sample ID	Value	Unit	Level	Assembly	Reference	RGD Study Report
SD-Ahrem1lexas+/+	male	Male	46 - 67 days	lung	GSM4947399	2064.69	TPM	high	GRCr8	RGD:401940192	GSE162239
SD-Ahrem1lexas+/+	male	Male	46 - 67 days	lung	GSM4947398	1920.79	TPM	high	GRCr8	RGD:401940192	GSE162239
SD-Ahrem1lexas+/+	male	Male	46 - 67 days	lung	GSM4947400	1860.17	TPM	high	GRCr8	RGD:401940192	GSE162239
SD-Il6em1Yona+/+	male	Male	49 - 63 days	lung	GSM8083919	1675.07	TPM	high	GRCr8	RGD:407450417	GSE255994
SD-Il6em1Yona-/-	male	Male	49 - 63 days	lung	GSM8083922	1611.76	TPM	high	GRCr8	RGD:407450417	GSE255994
SD-Il6em1Yona+/+	male	Male	49 - 63 days	lung	GSM8083918	1607.82	TPM	high	GRCr8	RGD:407450417	GSE255994
SD-Il6em1Yona-/-	male	Male	49 - 63 days	lung	GSM8083921	1539.97	TPM	high	GRCr8	RGD:407450417	GSE255994
SD-Il6em1Yona-/-	male	Male	49 - 63 days	lung	GSM8083923	1531.82	TPM	high	GRCr8	RGD:407450417	GSE255994

3C

Objective 3: Data Access in RGD – Expression Study Report

Expression Study: Aryl hydrocarbon receptor is essential for the pathogenesis of pulmonary arterial hypertension [day 4 RNA-Seq]

4A

Study ID:	3352
Study Name:	Aryl hydrocarbon receptor is essential for the pathogenesis of pulmonary arterial hypertension [day 4 RNA-Seq]
Study Source:	GEO
Data Type:	rna-seq_expression
Geo Series Acc:	GSE162239
Available Data:	Metadata and Reprocessed Data

Associated References

Aryl hydrocarbon receptor is essential for the pathogenesis of pulmonary arterial hypertension..
Masaki T, etal., Proc Natl Acad Sci U S A. 2021 Mar 16;118(11):e2023899118. doi: 10.1073/pnas.2023899118. PMID: 33836606, RGD ID: 401940192

Sample Metadata

4C

1 to 18 of 18 rows All Rows | Download Metadata | Column Selection ▲

Select Columns to Display: Select All | Clear All

GEO Accession, Tissue, Strain, Sex, Computed Sex, Age (in days), Life Stage, Experimental Conditions, Ordinality, Cell Type, Dose, Duration, Application Method, Notes

4B

geo_accession	Tissue	Cell Type	Strain	Sex	Computed Sex	Age(in days)	Life Stage	Experimental_Conditions	Ordinality	Dose	Duration	Application Method	Notes
GSM4947395	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	control condition	1				normoxia
GSM4947396	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	control condition	1				normoxia
GSM4947397	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	control condition	1				normoxia
GSM4947398	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	semaxanib	1	20.0 mg/kg		subcutaneous injection	
GSM4947398	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	controlled air oxygen content	2	10.0 %	4.0 days		
GSM4947399	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	semaxanib	1	20.0 mg/kg		subcutaneous injection	
GSM4947399	lung		SD-Ahrem1lexas+/+	male	Male	46-67	adult	controlled air oxygen content	2	10.0 %	4.0 days		

Figure 4. Expression Study Report page for GEO Series GSE162239.
A) Study summary with links to the original repository and publication(s).
B) Curated sample metadata, sortable by any column.
C) Column Selection customizes the displayed fields; all or selected records export to .csv.

The Data Downloads section (not shown) links to the study's download folder (see Figure 5).

GSE162239 Study Report

Objective 3: Data Access in RGD – Downloads & JBrowse2

RGD Expression Downloads

Index of /expression/GSE162239

Name	Last modified	Size	Description
Parent Directory		-	
Computed_attributes_README	2026-10-16 09:26	2.0K	
Expression_Methods_and_Disclaimer_README	2026-05-21 12:54	3.2K	
GSE162239.genes.TPM.matrix	2026-05-21 12:54	2.8M	
GSE162239.genes.counts.matrix	2026-05-21 12:54	3.1M	
GSE162239.transcripts.TPM.matrix	2026-05-21 12:54	8.7M	
GSE162239.transcripts.counts.matrix	2026-05-21 12:54	9.4M	
GSE162239_computed_sample_attributes_GRCr8.txt	2026-10-19 18:25	4.1K	
GSE162239_multiQC_report_GRCr8.html	2026-05-21 12:54	1.9M	
Genome-wide_read_coverage_BigWig_files/	2026-05-21 12:16	-	
mRatBN7_2_expression_data/	2026-05-21 12:16	-	

5A

Index of /expression/GSE162239 /Genome-wide_read_coverage_BigWig_files

Name	Last modified	Size	Description
Parent Directory		-	
GSE162239_jbrowse_session_GRCr8.json	2026-10-19 18:25	34K	
JBrowse2_README.pdf	2026-05-21 12:54	813K	
RNAseq_lung_SD-Ahrem1lexasMUT_M_GSM4947401.bigwig	2026-05-21 12:54	55M	
RNAseq_lung_SD-Ahrem1lexasMUT_M_GSM4947402.bigwig	2026-05-21 12:54	51M	
RNAseq_lung_SD-Ahrem1lexasMUT_M_GSM4947403.bigwig	2026-05-21 12:54	52M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947395.bigwig	2026-05-21 12:54	59M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947396.bigwig	2026-05-21 12:54	57M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947397.bigwig	2026-05-21 12:54	53M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947398.bigwig	2026-05-21 12:54	47M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947399.bigwig	2026-05-21 12:54	55M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947400.bigwig	2026-05-21 12:54	44M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947404.bigwig	2026-05-21 12:54	56M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947405.bigwig	2026-05-21 12:54	55M	
RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947406.bigwig	2026-05-21 12:54	51M	

5B

Figure 5. GSE162239 download directory. A) Gene- and transcript-level TPM and count matrices, computed sample attributes, and the MultiQC report.
B) Download a preconfigured session file and import it into RGD's JBrowse2 to load all of the study's coverage tracks. Then search for any gene to view its coverage (see Figure 6).

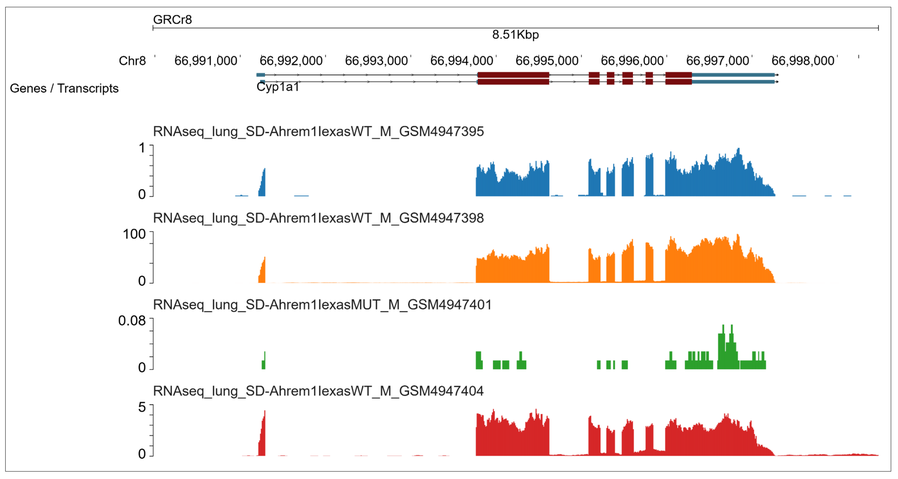


Figure 6. *Cyp1a1* coverage for GSE162239 in JBrowse2, shown after importing the session file (Figure 5B) and searching for *Cyp1a1* (exported PNG). BPM-normalized BigWig tracks, colored by experimental group (four groups, one representative sample per group is shown). Track heights are independently auto scaled by JBrowse2; interpretation of inferred gene expression should consider both the coverage profile and the displayed y-axis scale.

RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947395

METADATA	
Sample Characteristic	46.0-67.0d;control condition
Tissue	lung
Strain	SD-Ahrem1lexasWT
RGD Strain Report	https://rgd.mcw.edu/rgdweb/report/strain/main.html?id=401940198
Sex	M
Computed Sex	M
Computed Strandedness	Stranded (reverse)
Computed Library Capture	Poly(A) selection
RGD Metadata Report	https://rgd.mcw.edu/rgdweb/report/expressionStudy/main.html?geoAcc=GSE162239
Project Title	Aryl hydrocarbon receptor is essential for the pathogenesis of pulmonary arterial hypertension [day 4 RNA-Seq]
Project Repository Link	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE162239
Project Accession ID	GSE162239
Sample Accession ID	GSM4947395
PubMed ID	PMID:33836606
Data Processing	HPC RGD workflow; computed attributes added by RGD attribute backfill 1.0 (2026-10)
Read alignment	STAR v2.7.10b
Genome version	GCF_036323735.1 GRCr8
Expression Quantification	RSEM v1.3.1

RNAseq_lung_SD-Ahrem1lexasWT_M_GSM4947404

METADATA	
Sample Characteristic	lung endothelial cell;46.0-67.0d;control condition
Tissue	lung
Strain	SD-Ahrem1lexasWT
RGD Strain Report	https://rgd.mcw.edu/rgdweb/report/strain/main.html?id=401940198
Sex	M
Computed Sex	M
Computed Strandedness	Stranded (reverse)
Computed Library Capture	Poly(A) selection
RGD Metadata Report	https://rgd.mcw.edu/rgdweb/report/expressionStudy/main.html?geoAcc=GSE162239
Project Title	Aryl hydrocarbon receptor is essential for the pathogenesis of pulmonary arterial hypertension [day 4 RNA-Seq]
Project Repository Link	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE162239
Project Accession ID	GSE162239
Sample Accession ID	GSM4947404
PubMed ID	PMID:33836606
Data Processing	HPC RGD workflow; computed attributes added by RGD attribute backfill 1.0 (2026-10)
Read alignment	STAR v2.7.10b
Genome version	GCF_036323735.1 GRCr8
Expression Quantification	RSEM v1.3.1

Figure 7. "About Track" (from each track's menu) lists data processing details, computed sample attributes, and links to the RGD Expression Study Report, strain report, and original repository. "Sample Characteristic" shows the curated attributes that define each experimental group color in addition to the tissue_strain_sex track naming convention.

Conclusion & Future Directions

Metric	Value
Studies in the 2019 import (for comparison)	7 baseline studies
GEO Series reviewed and prioritized for curation	~3,000
Curated studies (baseline and experimental, in vivo)	>560
Unique curated samples	~11,000
Reprocessed Studies	540 (424 mRatBN7.2 and GRCr8; 116 GRCr8 only)
Tissue terms (UBERON)	206
Cell type (CL)	29
Rat strains (RS)	216 (including 49 newly registered strains)
Vertebrate trait (VT)	144
Clinical measurement (CMO)	161
Experimental condition (XCO)	696

Table 1. Growth of RGD expression content: GEO Series reviewed, curated studies and samples, reprocessed studies, rat strains, and distinct ontology terms.

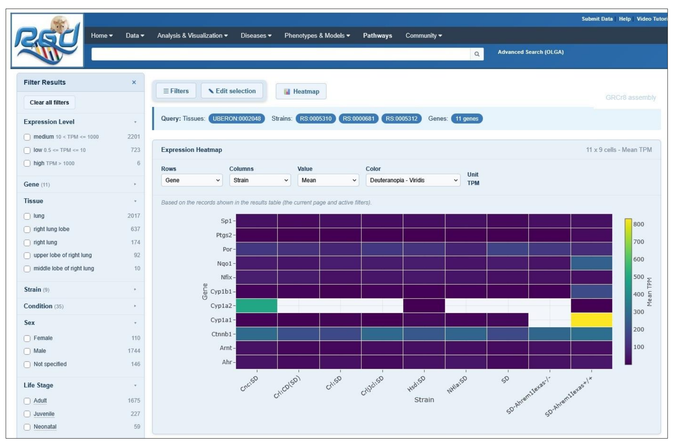


Figure 8. Mock-up of the Expression Portal, currently under development, showing a gene-by-strain heatmap for a user-supplied gene list.

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